



Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust - IERY)

TECHNICAL PROJECT REPORT

PROJECT TITLE

RA6M5 Processor Development Board

Domain of the Project

PCB Designing & Embedded Hardware

Team Mentor

Mr. Paramesh

(Mentor – SURE Trust)

Submitted By

Mr. L. Bharath Kumar

Project Duration

March 2026 – April 2026

DECLARATION

The project titled “**RA6M5 Processor Development Board**” has been successfully mentored by Mr. Paramesh and organized by **SURE Trust**, with the primary objective of empowering educated rural youth through hands-on industrial project training. This practical exposure bridges academic concepts with real-world employer requirements.

I hereby declare that the work presented in this technical report was carried out by the team member listed below, reflecting genuine effort, high standards of hardware design, and comprehensive technical learning.

Team Member Details:

- **Mr. L. Bharath Kumar**

Signature: _____

Mentor's Signature: _____

Name: Mr. Paramesh

Designation: Mentor – SURE Trust

Executive Director:

Prof. Radhakumari

SURE Trust

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1 Executive Summary

The primary objective of this project was to design a professional, high-performance embedded hardware development board centered around the Renesas RA6M5 microcontroller to support modern Internet of Things (IoT) applications. To achieve this, the board integrates multiple essential hardware modules onto a single printed circuit board (PCB), featuring a USB Type-C power input, battery charging and protection, Bluetooth Low Energy (BLE) communication, QSPI Flash memory, microSD card support, a Qwiic sensor connector, an SWD programming interface, status indicators, and multi-stage voltage regulation circuits.

Both the complete schematic capture and the professional four-layer PCB layout were meticulously engineered using KiCad software, adhering strictly to industry-standard PCB design practices, including precise component placement, controlled signal routing, clean power distribution, and rigorous Design Rule Check (DRC) verification. Ultimately, the project successfully consolidated complex peripheral requirements into a single compact and reliable development platform, eliminating the need for fragmented breakout modules and yielding a production-ready hardware layout that significantly enhanced practical competence in embedded systems architecture, hardware design, and electronic circuit development.

2 Introduction

Embedded systems form the foundation of almost every modern electronic device, ranging from smart home automation and industrial control systems to medical equipment and distributed Internet of Things (IoT) nodes. These systems demand reliable hardware capable of high-speed data processing, seamless external communication, and minimal power consumption. To gain direct industry-level experience and understand how such systems are developed, this project focuses on designing a complete embedded development board from scratch.

Modernizing hardware architectures is essential to meet the escalating performance demands of contemporary connected automation networks. Practical hardware realization bridges the gap between theoretical micro-controller concepts and actual industrial manufacturing workflows.

2.1 Problem Statement and Goals of the Project

Many modern IoT applications require several external modules, such as Bluetooth communication, memory storage, battery charging, and sensor interfaces. Designing and wiring all these modules separately increases system complexity, board size, and overall

cost. The primary goal of this project is to eliminate this complexity by integrating all required peripherals into a single, compact, and reliable printed circuit board (PCB).

Eliminating scattered evaluation modules minimizes parasitic noise paths and connection failures across sensitive signal lines. Consolidating core features provides a cost-effective and streamlined hardware baseline for multi-purpose sensing applications.

2.2 Scope and Limitations of the Project

The scope of this project encompasses complete schematic design, component selection, 4-layer PCB layout creation, power supply design, communication module integration, external memory support, and the preparation of manufacturing files such as Gerber and BOM. However, firmware programming and physical bench testing after PCB fabrication remain outside the scope of this hardware-centric project.

The layout scope covers multi-layer stackup design, rigorous design-rule checks, and component footprint verification. Post-fabrication hardware bring-up and embedded code development are explicitly excluded from the current design cycle deliverables.

2.3 Innovation Component in the Project

Unlike simple standard evaluation boards, this unique design integrates multiple industrial-level features onto a single board. These include a USB Type-C power input, an autonomous battery charging circuit, a battery fuel gauge, Bluetooth Low Energy (BLE) communication, external QSPI Flash memory, microSD card storage, a Qwiic sensor connector, and an SWD programming interface. This comprehensive integration makes the development board highly suitable for advanced, real-world embedded and IoT applications.

Incorporating a dedicated hardware fuel gauge allows precise state-of-charge tracking for standalone field deployments. The addition of a standardized Qwiic connector enables rapid, tool-free expansion using external modular environmental sensors.

3 Project Objectives and Methodology

3.1 Clearly Defined Objectives and Goals of the Project

The main objectives of this project are to design a complete embedded development board based on the Renesas RA6M5 microcontroller and to create the complete circuit schematic using KiCad. The goals include designing a professional 4-layer PCB following industry-standard PCB design practices, and integrating essential peripherals such as USB Type-C, Bluetooth Low Energy (BLE), QSPI Flash memory, a microSD card slot, and a Qwiic connector into a single board. Additionally, the project aims to design

a reliable power management system equipped with battery charging and voltage regulation, ensure design error-free by performing Electrical Rule Check (ERC) and Design Rule Check (DRC), generate manufacturing outputs like Gerber files and Bills of Materials (BOM) for fabrication, and improve practical knowledge and hands-on experience in embedded hardware and PCB development.

3.2 Expected Outcomes and Deliverables

The expected outcomes of this project include a complete and error-free schematic capture along with a professional 4-layer PCB layout designed in KiCad. The key deliverables consist of manufacturing-ready Gerber and BOM files, a compact development board tailored for embedded and IoT applications, and significantly enhanced technical skills in PCB design, component selection, and hardware development.

3.3 Methods and Tools Used

- **Methods/Technology used:** The project was carried out using a systematic approach to design a professional embedded development board, ensuring the final PCB design was reliable, manufacturable, and suitable for IoT and embedded system applications. The design process followed a structured multi-step methodology, including requirement analysis, component selection, schematic design with electrical checks, 4-layer PCB layout routing, design rule verification, and manufacturing output generation.
- **Tools/Software used:** The project was designed using KiCad, an open-source Electronic Design Automation (EDA) software. KiCad served as the primary environment for schematic design, PCB layout, design verification (ERC and DRC), 3D visualization, and generating Gerber and BOM files required for PCB manufacturing.
- **Data collection approach:** Component parameters, electrical specifications, and functional layout requirements were gathered from industry-standard datasheets and application notes to ensure proper compatibility, performance, and hardware reliability.

4 Project Architecture and Working Screenshots Explanation

4.1 Project Architecture & Flow Diagrams

The system follows a structured modular architecture. Below are three different visual representations showcasing the power distribution, core microcontroller processing, and peripheral subsystem integration.

4.1.1 Flowchart Style 1: Graphical Block Architecture (TikZ)

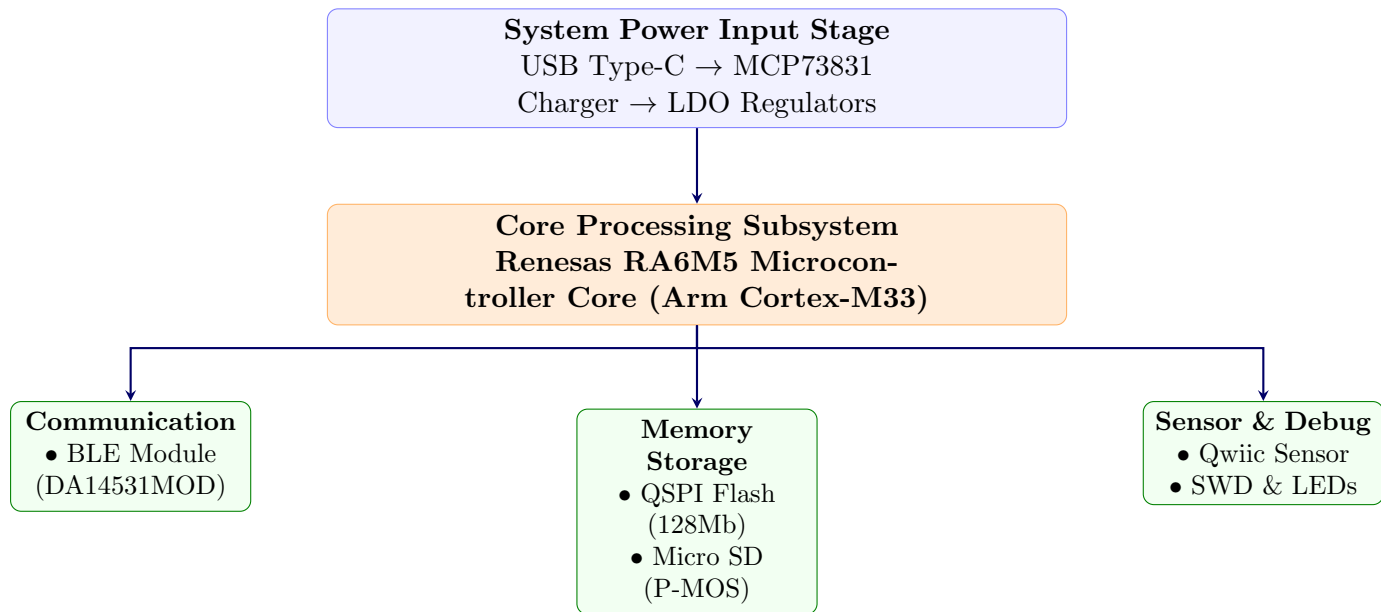


Figure 1: System Architecture Flowchart - Style 1 (Graphical Blocks)

4.1.2 Flowchart Style 2: Structured Box Architecture (TikZ Layout)

4.1.3 Flowchart Style 3: Sequential Linear Pipeline Architecture

- **Power Flow & Regulation:** External power is supplied via the USB Type-C connector, which feeds the battery charging circuitry and regulators to provide steady voltage rails across the board.
- **Core Processing Unit:** The Renesas RA6M5 microcontroller serves as the central intelligence, coordinating all internal data operations and peripheral interfaces.
- **Peripheral Subsystem Distribution:** The architecture groups external components logically into communication links, memory storage arrays, and expansion or debugging interfaces to ensure clean and organized hardware integration.

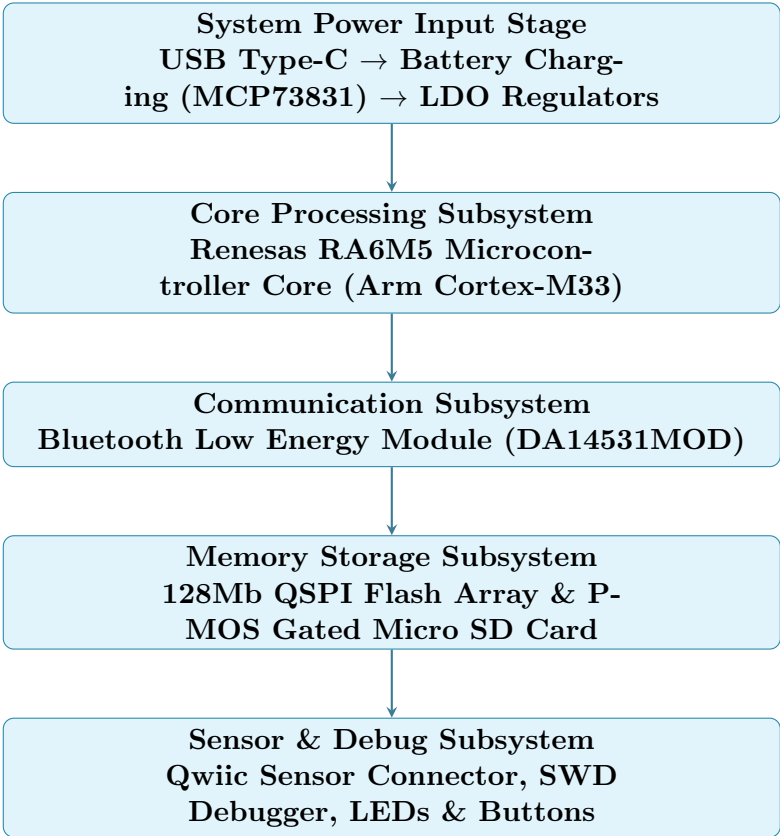


Figure 2: System Architecture Flowchart - Style 2 (Structured Box Layout)

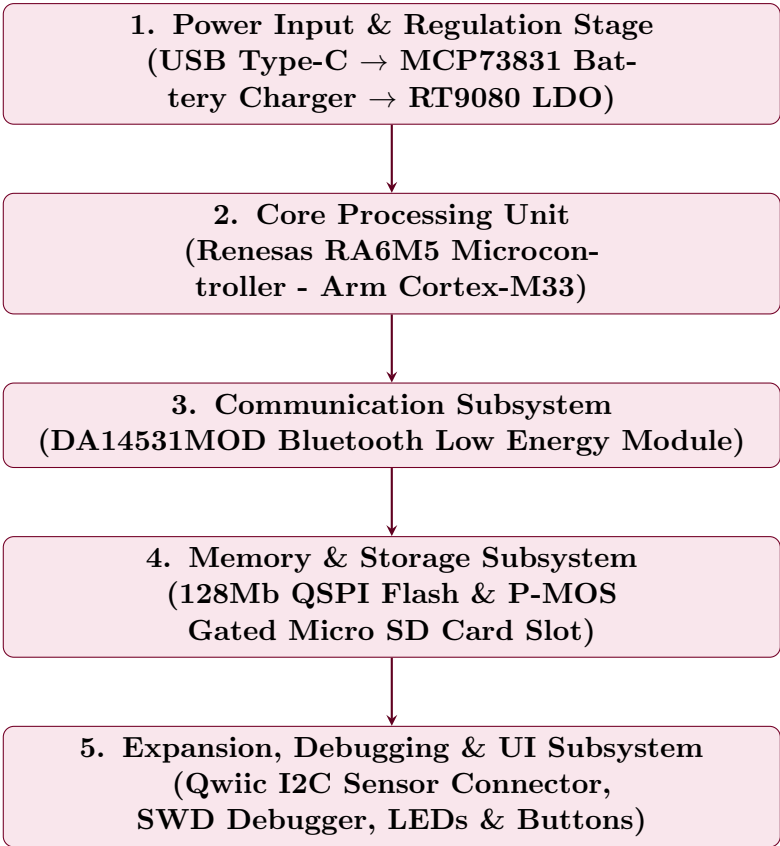


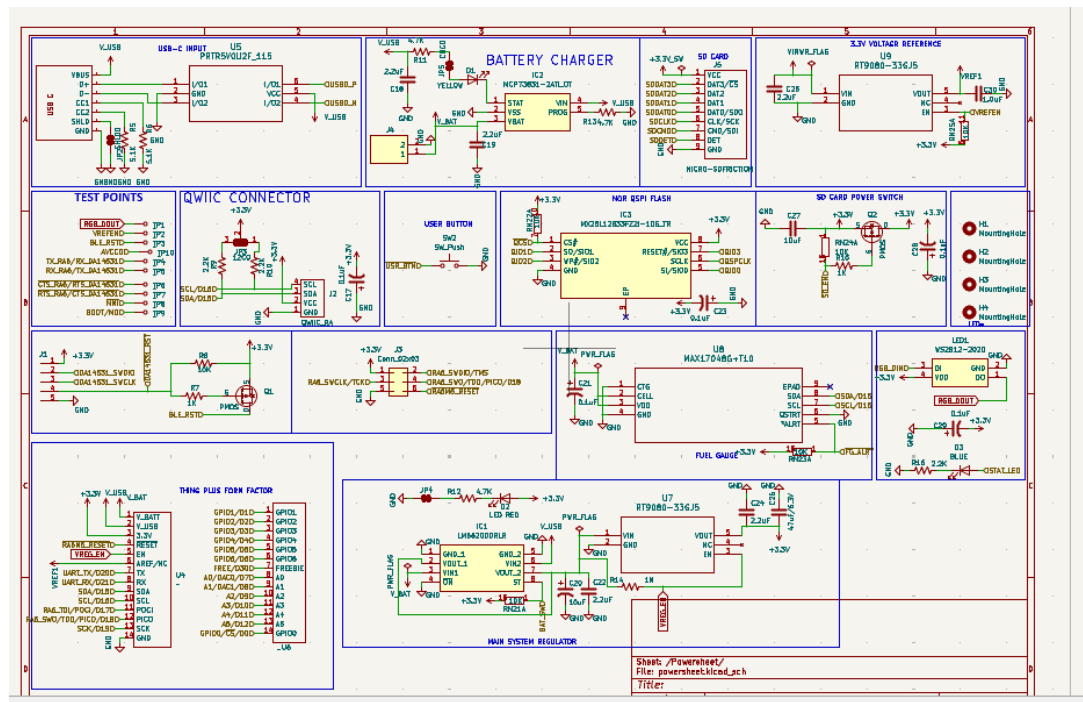
Figure 3: System Architecture Flowchart - Style 3 (Sequential Pipeline)

4.2 Final Project Working Screenshots & Visual Verification

4.2.1 Complete Schematic Diagram

The hierarchical schematic captures all functional blocks—ranging from the power regulation sheet to the main controller sheet—verified thoroughly through Electrical Rule Checks (ERC) to ensure zero open nodes or connection conflicts.

- The power supply and regulation sheet details the USB-C input stage, ESD protection, LM66200/RT9080 regulators, MCP73831 battery charger, and MAX17048 fuel gauge circuits.
- The main controller sheet captures the Renesas RA6M5 microcontroller core, digital GPIO routing, analog input conditioning, crystal oscillators, and communication headers.
- Root hierarchical sheet interconnections establish the top-level link between the modular power management sub-sheet and main processor sheet via standardized bus lines.



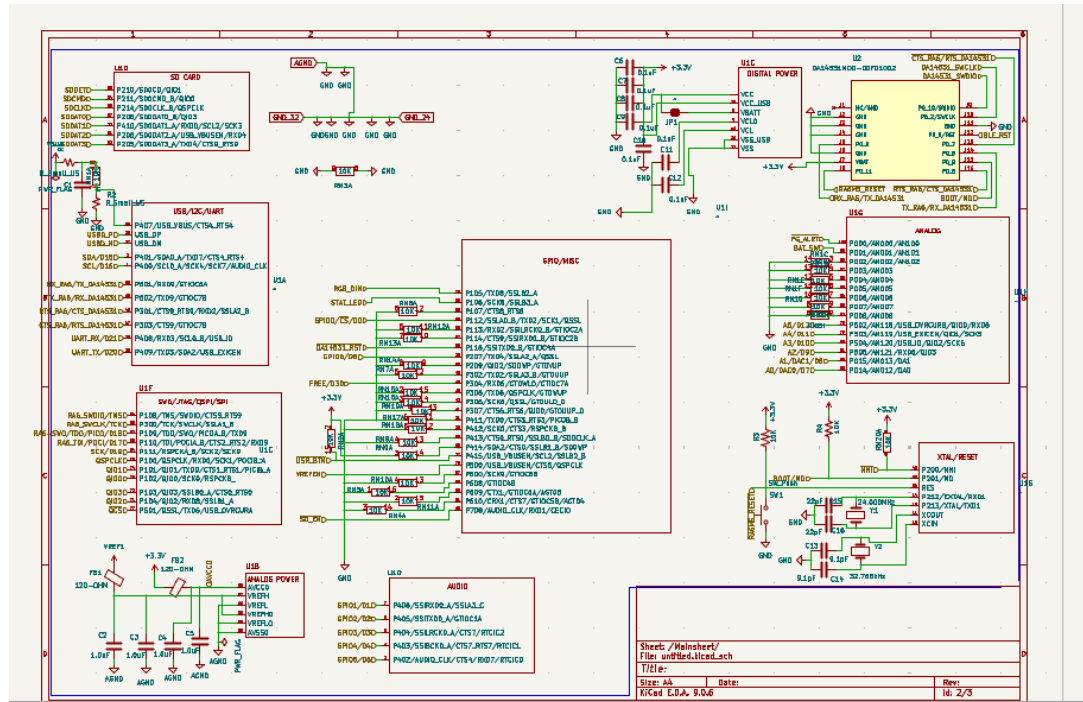


Figure 5: Main Microcontroller and Peripherals Schematic Sheet

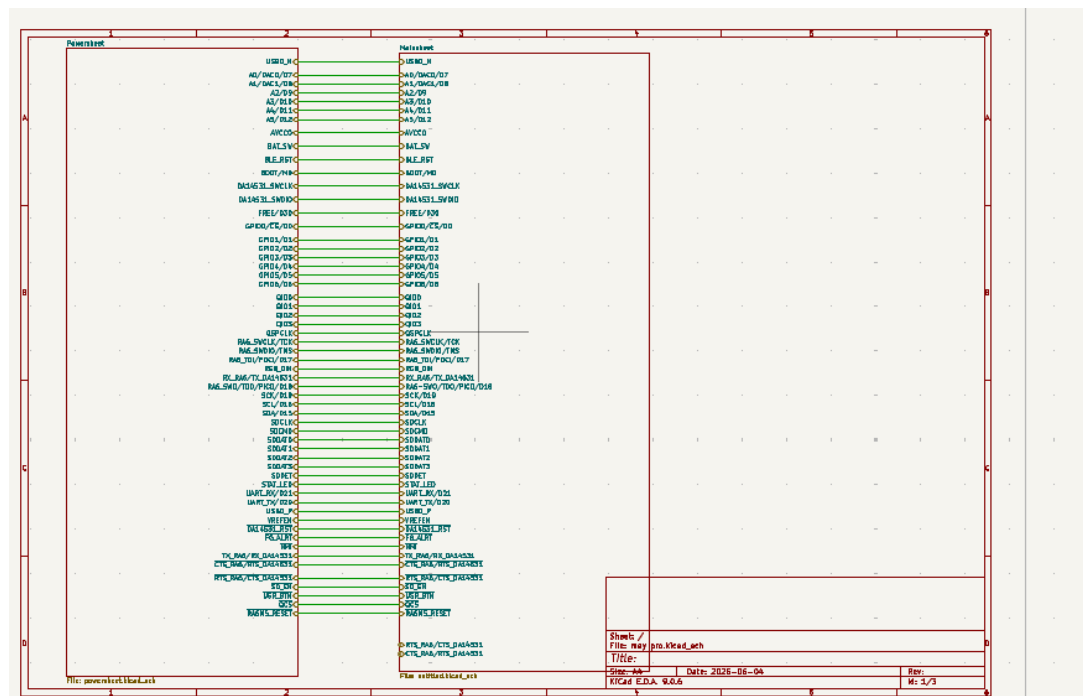


Figure 6: Root Hierarchical Interconnections Schematic Sheet

4.2.2 PCB Layout

The multi-layer board view illustrates the precise 4-layer stackup implementation, showing how component placement, high-speed signal routing on external copper layers, in-

ternal power planes, and dedicated ground pours were executed in KiCad and validated via Design Rule Checks (DRC).

- The combined multi-layer routing layout view displays top-layer traces, internal power planes, bottom copper ground pours, and design rule compliance across the board.
- Dedicated front assembly and top trace configurations optimize high-speed signal paths while maintaining safe component clearances.
- Bottom-layer trace routing paths, vias, and continuous ground plane coverage provide effective noise shielding and stable power return paths.

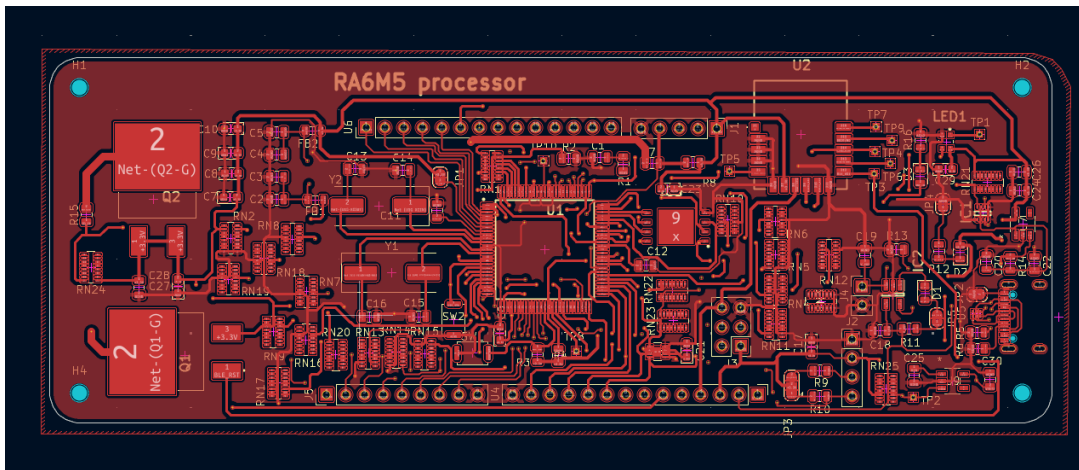


Figure 7: PCB Layout - Top Copper Layer Routing and Placement

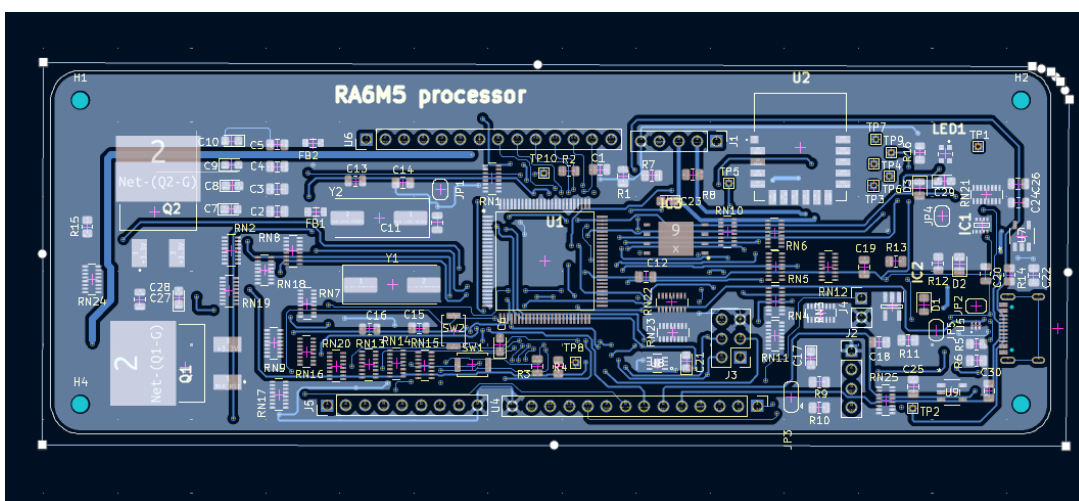


Figure 8: PCB Layout - Bottom Copper Layer Routing and Traces

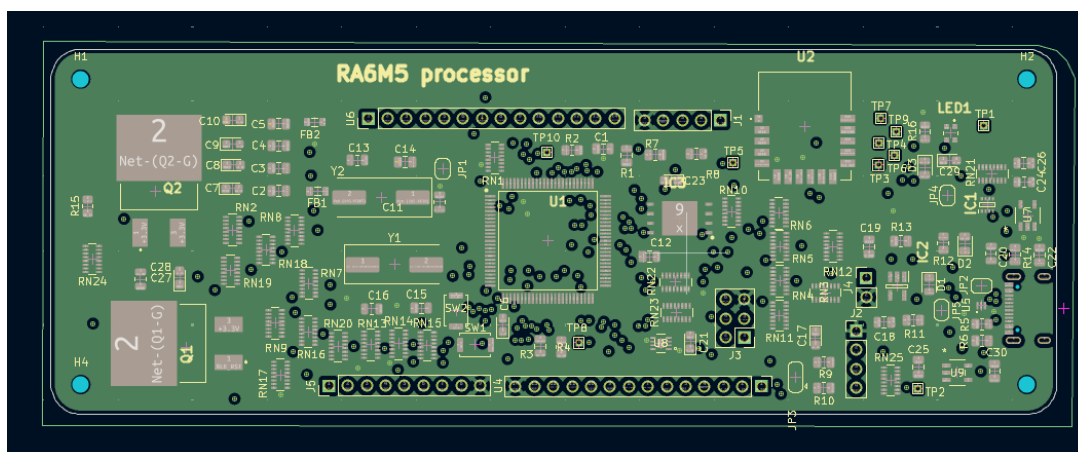


Figure 9: PCB Layout - Internal Ground Plane View

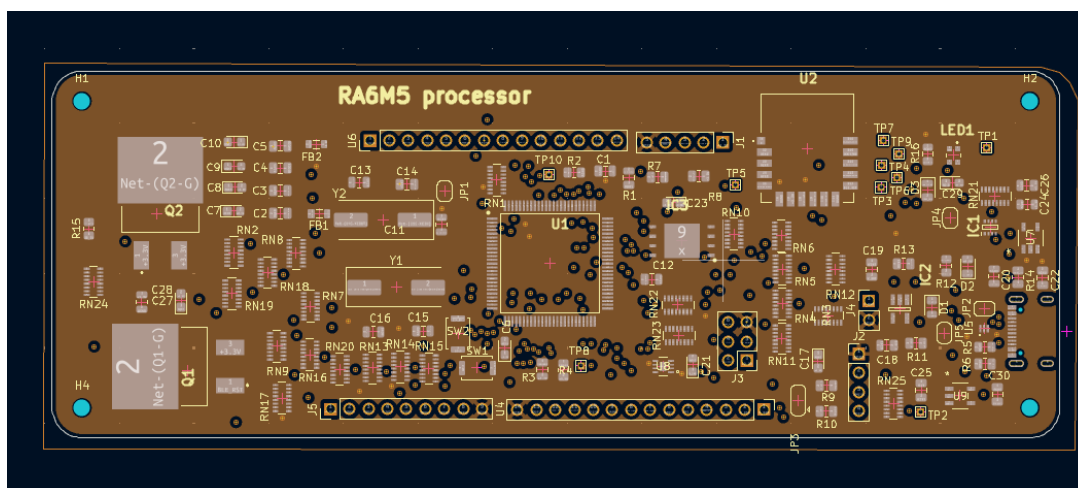


Figure 10: PCB Layout - Internal Power Plane View

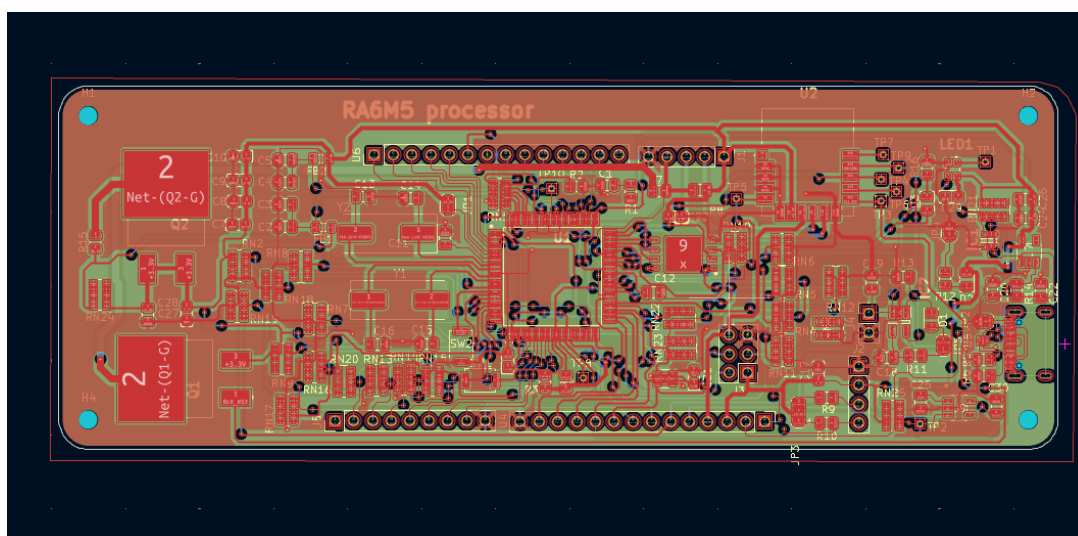


Figure 11: PCB Layout - Combined Multi-Layer Routing View

4.2.3 3D View of the PCB

The realistic 3D rendered view provides physical verification of the assembled board, confirming proper component heights, correct connector orientations, silkscreen readability, and mechanical clearance for the mounting holes.

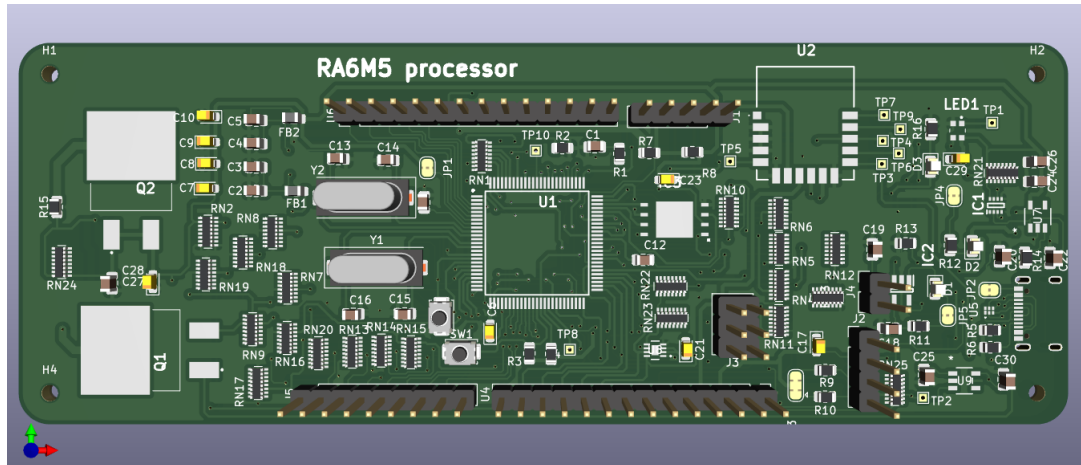


Figure 12: 3D Rendered View of the PCB - Top Assembly and Component Placement

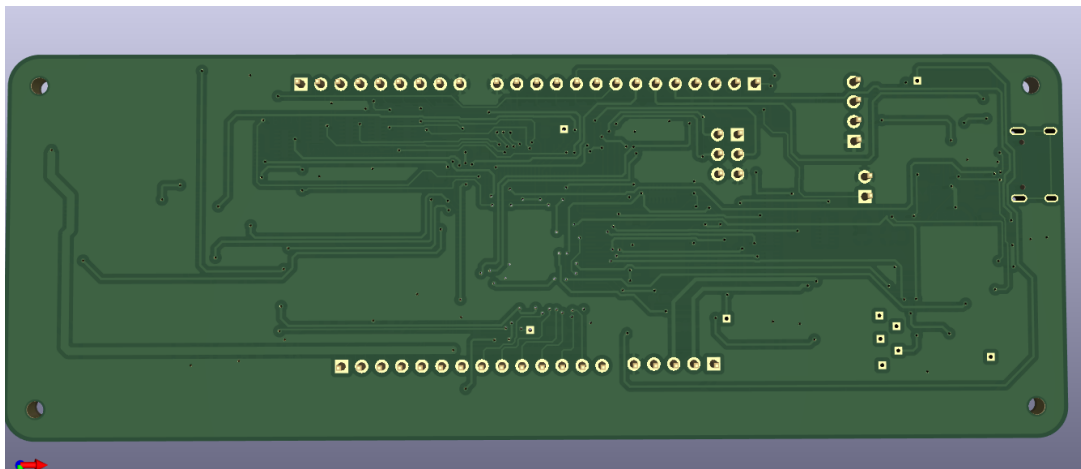


Figure 13: 3D Rendered View of the PCB - Bottom View and Trace Routing Verification

5 Learning and Reflection

5.1 Document Every Team Member's New Learnings

- **Technical Learnings in Hardware Design:** Acquired deep practical knowledge in embedded hardware architecture, board-level component selection, and multi-sheet schematic capture using KiCad EDA.

- **PCB Stackup Routing Competence:** Mastered the execution of a professional 4-layer PCB layout, implementing clean power distribution planes, ground pours, and strict signal routing practices to minimize noise interference.
- **Peripheral Subsystem Integration:** Learned how to effectively interface and route complex components, including USB-C power delivery circuits, battery chargers, Bluetooth Low Energy (BLE) modules, QSPI Flash memory, microSD slots, and Qwiic sensor connectors.
- **Design Validation Manufacturing Outputs:** Gained hands-on proficiency in performing thorough Electrical Rule Checks (ERC) and Design Rule Checks (DRC), alongside generating fabrication-ready Gerber files and Bills of Materials (BOM).

5.2 Document Every Team Member's Experience

- **End-to-End Hardware Journey:** Working on this project provided an invaluable, hands-on experience in taking an advanced IoT development board from initial conceptualization all the way to a manufacturing-ready layout.
- **Overcoming Technical Hurdles:** Initially, challenges arose during component footprint selection, high-speed trace routing, and resolving ERC/DRC verification warnings; however, these obstacles were systematically overcome through rigorous datasheet analysis and mentor guidance.
- **Professional Growth Confidence:** The overall experience substantially strengthened technical confidence, analytical problem-solving abilities, and a comprehensive understanding of the hardware design workflows utilized in the modern electronics industry.

6 Conclusion and Future Scope

6.1 Recap Objectives and Achievements

The primary objective of designing a professional, high-performance embedded hardware development board centered around the Renesas RA6M5 microcontroller was successfully achieved. The project milestones were fully realized through meticulous multi-sheet schematic capture and a robust 4-layer PCB layout designed entirely in KiCad. Essential subsystems—including USB Type-C power input, autonomous battery charging, Bluetooth Low Energy (BLE) communication, external QSPI Flash memory, microSD card storage, a Qwiic sensor connector, and SWD debugging interfaces—were seamlessly integrated onto a single compact platform. Rigorous verification through Electrical Rule

Checks (ERC) and Design Rule Checks (DRC) ensured that the hardware architecture is entirely error-free, structurally sound, and fully prepared for industrial manufacturing. Ultimately, this project successfully bridged theoretical engineering concepts with practical hardware design workflows.

6.2 Future Scope of this Project

- **Embedded Firmware Development:** Writing, compiling, and deploying dedicated peripheral firmware and bare-metal drivers for the Renesas RA6M5 microcontroller to execute real-world tasks.
- **Wireless Expansion:** Integrating supplementary wireless communication protocols, such as Wi-Fi or cellular telemetry, to broaden connectivity options for distributed industrial environments.
- **Sensor Array Scalability:** Expanding environmental and industrial sensing capabilities by connecting multiple modular sensors via the onboard Qwiic connector interface.
- **Cloud IoT Integration:** Establishing secure MQTT or HTTP telemetry pipelines to stream live sensor data from the development board to cloud-based monitoring dashboards.
- **Power Optimization:** Fine-tuning low-power sleep modes and energy-harvesting profiles to maximize battery operational lifetime for remote field deployments.
- **Physical Prototyping Testing:** Fabricating physical PCB prototypes and conducting exhaustive bench testing using oscilloscopes and logic analyzers to validate real-world signal integrity and electrical performance.
- **Custom Enclosure Design:** Designing and 3D-printing a rugged, protective housing tailored to safeguard the development board during harsh field applications.